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A practical perspective on the drone-with-a-slung-load problem

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Abstract— **Controlling a system consisting of a drone carrying a payload with a cable is a problem of practical importance within UAV control. Contrarily to previous work, this paper investigates a control scheme for this problem where neither the deviation angles of the cable nor the mass of the load are known or measured, with an emphasis on simplicity and applicability on a wide array of available flight controllers. Our approach combines differential-algebraic considerations for motion planning and trajectory generation, together with simple controllers used for feedback. Simulation results are proposed to illustrate the potential of the approach.**

I. Introduction

In the area of UAV control, control of multirotors with a load suspended by a cable, also referred to as slung-load has gained significant popularity over the past few years [1]–[4]. Regarding control design, it is indeed an interesting problem, the system under consideration being nonlinear, underactuated, and presenting potentially challenging oscillations. But it is also practically relevant: in terms of mechanism, it is indeed generally easy or convenient to attach a load to a cable suspended under a drone using simple attachments and hooks, especially when compared to having a special device to place/retain a payload rigidly under a multirotor drone.

Among the different approaches used to address this control problem, differential flatness has gained significant attention [5], [6]. Seen as an extension of controllability for nonlinear systems, flatness allows the parametrization of a system state and inputs using the so-called *flat outputs* and a finite number of their derivatives. One of the many advantages of such a perspective is to allow one to perform motion planning in a simple manner, without solving differential equations.

However, the vast majority of the above-cited studies use the knowledge of the mass of the load, as well as measurement of the angles between the load and the drone as starting assumptions in order to solve the control problem. With regard to actual drone operations, this is rarely the case in practice. Indeed, the load mass is meant to change as per the requirements of the mission and is not always known in advance, while measuring the deflection angles of the cable for an outdoor operation requires a sensing mechanism attached under the drone, typically an unactuated gimbal equipped with encoders. Nonetheless, many drones transporting loads with cables do not currently use such mechanisms (see for example [7]).

While a somewhat straight approach to tackle a lack of information regarding the load mass and its deflection angles would consist in an estimation scheme for all missing measurements, doing so is not with different potential pitfalls such as difficulty in design, analysis and, especially important for practical reasons, tuning. In order to retain the advantages and performances of differential flatness, this paper will use the latter technique, but taking a slightly different perspective than current studies on drone control. Indeed, taking as the starting point earlier references on crane control using flatness (see [8], [9] and [10], we only perform the nonlinear controllability analysis and subsequent motion planning on a subpart or projection of the drone-slung-load dynamical model not directly involving the load mass nor the deflection angles. This is especially useful for rest-to-rest trajectories that are of primary importance with regard to practice in operational scenarios. Once the main part of control decisions is taken care of by motion planning, model inaccuracies and kinetics (including the unknown load mass) can be compensated for by a simple feedback controller, here materialized by a well-known active disturbance rejection controller (ADRC).

The rest of this paper is organized as follows. Section II briefly introduces and motivates the drone-with-a-slung-load model. Then, Section III considers a selection of a part of this model and proves its flatness property. In Section IV, we present the control architecture of our approach, which includes motion planning and feedback control, with a focus on simplicity and practicality, while Section V illustrates the approach with a few simulation results. Brief concluding remarks end this paper.

II. Modelling

Consider a drone carrying a payload by an attached cable as shown in Figure 1. The variables used in this paper are listed in Table I.

In the following, we make the following assumptions about the system (see also [11] for similar assumptions):

- The attached cable is mass-less and taut and installed at the mass center of the drone.
- The payload is always situated under the drone, i.e. the swing angles α and β are within the range of $(-\pi/2, \pi/2)$.
- The drone and the payload are both assumed to be point masses, and the effect of aerodynamics on the drone, cable, and payload are neglected.

Note that the cable suspending the load being attached to the center of the gravity (COG) of the drone, the payload will not have any effect on the rotational motion of the drone [12].

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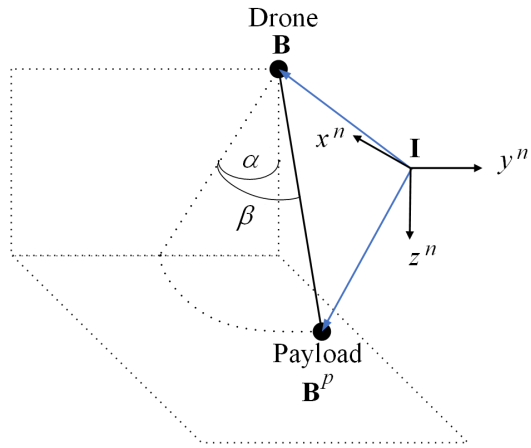

Fig. 1: Pendulum angle parametrization for the drone with cable-suspended cable

TABLE I: Symbol Definitions

Symbol	Definition
$m, m_p \in \mathbb{R}$	Mass of the drone and the payload
$g \in \mathbb{R}$	Gravitational constant
I	Inertial frame
B^p, B	Payload and drone body frame
$\mathbf{p} = [x, y, z]^\top \in \mathbb{R}^3$	Position of the drone in I
$\mathbf{p}^p = [x^p, y^p, z^p]^\top \in \mathbb{R}^3$	Position of the payload in I
$\alpha, \beta \in \mathbb{R}$	Swing angles of the payload
$l \in \mathbb{R}$	Cable length
$\boldsymbol{\eta} = [\mathbf{p}^\top, \alpha, \beta]^\top \in \mathbb{R}^5$	Generalized coordinates vector
$\mathbf{R} \in SO(3)$	Rotation matrix from B to I
$f \in \mathbb{R}$	Thrust directed along negative z-axis in B
$\mathbf{F} = [F_1, F_2, F_3]^\top \in \mathbb{R}^3$	Inertial control force
$\mathbf{w} = [w_1, w_2, w_3]^\top \in \mathbb{R}^3$	Additional environment forces and disturbances
$P, K \in \mathbb{R}$	Potential and kinetic energy
$\mathbf{M}(\boldsymbol{\eta}) \in \mathbb{R}^{5\times5}$	Mass matrix
$\mathbf{C}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) \in \mathbb{R}^{5\times5}$	Coriolis matrix
$\mathbf{G}(\boldsymbol{\eta}) \in \mathbb{R}^5$	Gravity vector
$M_{ij}, C_{ij}, G_{ij} \in \mathbb{R}$	An array of matrix with i and j as the row and column
$T \in \mathbb{R}$	Tension on the cable
$\mathbf{u}_{FF}, \mathbf{u}_{FB} \in \mathbb{R}^3$	Feedforward and feedback control
$\mathbf{p}_d = [x_d, y_d, z_d]^\top \in \mathbb{R}^3$	Desired position of the drone
$c_d, \rho \in \mathbb{R}$	Drag coefficient, air density
$\mathbf{v}_r, \mathbf{v}_{wind} \in \mathbb{R}^3$	Relative and wind velocity
$\mathbf{A} \in \mathbb{R}^3$	Reference area
$l_x, l_y, l_z \in \mathbb{R}$	Length, width and height of drone

The geometric constraint between the drone and payload positions (see Figure 1) is expressed by

$$\mathbf{p}^p = \mathbf{p} + \mathbf{L} \tag{1}$$

with

$$\mathbf{L} = \begin{bmatrix} \sin\beta \\ -\sin\alpha\cos\beta \\ \cos\alpha\cos\beta \end{bmatrix} l, \tag{2}$$

where l is the length of the cable, and α, β are the swing angles of the payload.

A dynamical model of the drone-payload system can be obtained by the Lagrangian method [13]. Considering $\boldsymbol{\eta} = [\mathbf{p}^\top, \alpha, \beta]^\top$, the Euler-Lagrange equations with respect to $\boldsymbol{\eta}$ are given by

$$\frac{d}{dt}\left(\frac{\partial E}{\partial \boldsymbol{\eta}}\right) - \frac{\partial E}{\partial \dot{\boldsymbol{\eta}}} = \boldsymbol{\tau} + \boldsymbol{\tau_d}, \tag{3}$$

where the Lagrangian is $E = K - P$, the generalized forces vector is $\boldsymbol{\tau} = [\mathbf{F}^\top, \mathbf{0}_{1\times2}]^\top$ and $\mathbf{F} = f\mathbf{R}\mathbf{e}_z$ with $\mathbf{e}_z = [e_{z,1}, e_{z,2}, e_{z,3}] = [0, 0, 1]^\top$. The additional environmental forces disturbances such as wind are shown in the $\boldsymbol{\tau_d} = [\mathbf{w}^\top, \mathbf{0}_{1\times2}]^\top$. The potential and kinetic energy of the drone-payload system are

$$P = -g(mz + m_p z^p) \tag{4}$$

$$K = \frac{1}{2}\left(m(\dot{\mathbf{p}})^\top\mathbf{p} + m_p(\dot{\mathbf{p}}^p)^\top\mathbf{p}^p\right) \tag{5}$$

so that we have the dynamical model

$$\mathbf{M}(\boldsymbol{\eta})\ddot{\boldsymbol{\eta}} + \mathbf{C}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}})\dot{\boldsymbol{\eta}} + \mathbf{G}(\boldsymbol{\eta}) = \boldsymbol{\tau} + \boldsymbol{\tau_d} \tag{6}$$

where the elements of the mass matrix $\mathbf{M}$ are

$$\begin{cases} M_{11} = M_{22} = M_{33} = m + m_p, \\ M_{44} = m_p l^2\cos^2(\beta) \\ M_{55} = m_p l^2 \\ M_{15} = M_{51} = m_p l\cos(\beta) \\ M_{24} = M_{42} = -m_p l\cos(\alpha)\cos(\beta) \\ M_{25} = M_{52} = m_p l\sin(\alpha)\sin(\beta) \\ M_{12} = M_{13} = M_{14} = M_{21} = M_{23} = 0 \\ M_{31} = M_{32} = M_{41} = M_{45} = M_{54} = 0 \end{cases} \tag{7}$$

while the elements of the Coriolis matrix $\mathbf{C}$ are

$$\begin{cases} C_{ij} = 0, \quad \text{for} \quad i = 1,2,3 \quad \text{and} \quad j = 1,2,\ldots,5 \\ C_{14} = C_{55} = 0 \\ C_{15} = -m_p l\dot{\beta}\sin(\beta) \\ C_{24} = m_p l(\dot{\alpha}\sin(\alpha)\cos(\beta) + \dot{\beta}\sin(\beta)\cos(\alpha)) \\ C_{25} = m_p l(\dot{\alpha}\sin(\beta)\cos(\alpha) + \dot{\beta}\sin(\alpha)\cos(\beta)) \\ C_{34} = m_p l(\dot{\beta}\sin(\alpha)\sin(\beta) - \dot{\alpha}\cos(\beta)\cos(\alpha)) \\ C_{35} = m_p l(\dot{\alpha}\sin(\alpha)\sin(\beta) - \dot{\beta}\cos(\alpha)\cos(\beta)) \\ C_{44} = -m_p l^2(\dot{\beta}\cos(\beta)\sin(\beta) \\ C_{45} = -C_{54} = m_p l^2\dot{\alpha}\cos(\beta)\sin(\beta) \end{cases} \tag{8}$$

and the elements of the gravity vector $\mathbf{G}$ are given by

$$\begin{cases} G_{11} = G_{21} = 0 \\ G_{31} = -(m_p + m)g \\ G_{41} = m_p g l\cos(\beta)\sin(\alpha) \\ G_{51} = m_p g l\cos(\alpha)\sin(\beta) \end{cases}. \tag{9}$$

Since the mass matrix $\mathbf{M}(\boldsymbol{\eta})$ is invertible, a state-space form of dynamical model (6) can easily be obtained.

Note that there are alternative ways to describe the above drone-payload system. Indeed, a straightforward and equivalent Newtonian approach would introduce a force representing the tension in the cable, so that we would have

$$m\ddot{\mathbf{p}} = \mathbf{F} + mg\mathbf{e}_z + \mathbf{f}_T + \mathbf{w}, \tag{10}$$

where the force vector along the cable $\mathbf{f}_T$ is given by

$$\mathbf{f}_T = \begin{bmatrix} \sin\beta \\ -\sin\alpha\cos\beta \\ \cos\alpha\cos\beta \end{bmatrix} T. \tag{11}$$

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Then, the payload dynamics are simply expressed as

$$m_p \ddot{\mathbf{p}}^p = -\mathbf{f}_T + m_p g \mathbf{e}_z. \tag{12}$$

III. Differential Flatness

This section briefly recalls the definition of differential flatness and proves the flatness property of a subset of the drone-with-a-slung-load system.

A system is differentially flat if one can find a set of outputs such as the states and the inputs of the system can be written as functions of these outputs. Following [14], the definition of a flat system is given as follows.

Definition 1: A nonlinear system

$$\dot{\mathbf{x}} = \mathbf{g}(\mathbf{x}, \mathbf{u}) \tag{13}$$

where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{u} \in \mathbb{R}^m$, is differentially flat if there is a vector $\boldsymbol{\sigma}$ called flat output, consisting of m differentially independent variables functions of the state, the input, and its derivatives, ie we have

$$\boldsymbol{\sigma} = \boldsymbol{\zeta}(\mathbf{x}, \mathbf{u}, \dot{\mathbf{u}}, ..., \mathbf{u}^{(s-1)}) \tag{14}$$

such that the state and the input vectors are functions of the flat output and its derivatives, ie we have

$$\mathbf{x} = \boldsymbol{\mu}(\boldsymbol{\sigma}, \dot{\boldsymbol{\sigma}}, \ddot{\boldsymbol{\sigma}}, \ldots, \boldsymbol{\sigma}^{(q-1)}) \tag{15}$$

and

$$\mathbf{u} = \boldsymbol{\nu}(\boldsymbol{\sigma}, \dot{\boldsymbol{\sigma}}, \ddot{\boldsymbol{\sigma}}, \ldots, \boldsymbol{\sigma}^{(q)}), \tag{16}$$

where $\boldsymbol{\alpha}, \boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are vectors of smooth functions and s and q are finite indices.

In the following, we will focus on the differential-algebraic equations (DAEs) describing the pendulum behavior of the drone-payload system represented in (1)-(2) and (11)-(12) and prove its flatness property. Note that, in doing so, the variable $\mathbf{p}$ representing the drone position is here regarded as an input.

Theorem 1: The system consisting of (1)-(2) and (11)-(12) with dynamic variables $\mathbf{p}^p, T, \alpha, \beta$ and input $\mathbf{p}$ is differentially flat, with $\boldsymbol{\sigma} = [\sigma_x, \sigma_y, \sigma_z]^\top = \mathbf{p}^p = [x^p, y^p, z^p]^\top$ as flat output.

Proof: Rewriting DAEs (11)-(12) using flat output(s) $\boldsymbol{\sigma} = \mathbf{p}^p$ gives

$$\begin{aligned} m_p \ddot{\sigma}_x &= -T \sin\beta && (17)\\ m_p \ddot{\sigma}_y &= T \sin\alpha \cos\beta && (18)\\ m_p(\ddot{\sigma}_z - g) &= -T \cos\alpha \cos\beta. && (19) \end{aligned}$$

To obtain T as a function of $\boldsymbol{\sigma}$ and its derivatives, square (17),(18),(19) and add them together to get

$$m_p^2 \left(\ddot{\sigma}_x^2 + \ddot{\sigma}_y^2 + (\ddot{\sigma}_z - g)^2\right) = T^2 \tag{20}$$

whereby, since cable tension $T \geq 0$ by assumption, T is readily obtained after a square root. Then, divide (18) by (19) to get

$$\tan\alpha = \frac{\ddot{\sigma}_y}{g - \ddot{\sigma}_z}, \tag{21}$$

so that α is computed thanks to the $\arctan$ function. Having equation (21) to the corresponding sine function expression

$$\sin\alpha = \frac{\ddot{\sigma}_y}{\sqrt{\ddot{\sigma}_y^2 + (\ddot{\sigma}_z - g)^2}}. \tag{22}$$

Replace $\sin\alpha$ in equation (18) using (22), and divide eq. (17) by eq. (18) to get

$$\tan\beta = -\frac{\ddot{\sigma}_x}{\sqrt{\ddot{\sigma}_y^2 + (\ddot{\sigma}_z - g)^2}} \tag{23}$$

so that we have β as functin of $\boldsymbol{\sigma}$ and derivatives.
Then, rewrite equations (1)-(2) as

$$\begin{aligned} \sigma_x - x &= l \sin\beta && (24)\\ \sigma_y - y &= -l \sin\alpha \cos\beta && (25)\\ \sigma_z - z &= l \cos\alpha \cos\beta. && (26) \end{aligned}$$

Notice that eq. (19) multiplied by eq. (25) is equal to eq. (18) multiplied by eq. (26), while eq. (19) multiplied by eq. (24) is equal to eq. (17) multiplied by eq. (26), so that we have

$$\begin{aligned} (\ddot{\sigma}_z - g)(\sigma_y - y) &= \ddot{\sigma}_y(\sigma_z - z) && (27)\\ (\ddot{\sigma}_z - g)(\sigma_x - x) &= \ddot{\sigma}_x(\sigma_z - z). && (28) \end{aligned}$$

Squaring all (24)-(26) equations and add them to get the obvious expression

$$(\sigma_x - x)^2 + (\sigma_y - y)^2 + (\sigma_z - z)^2 = l^2 \tag{29}$$

from which it can easily be inferred that z is a function of corresponding to the length constraint on the cable. Isolating $\sigma_y - y$ in (27) and $\sigma_x - x$ in (28), use these expressions in (29) to obtain

$$\sigma_z - z = \frac{l(\ddot{\sigma}_z - g)}{\sqrt{\ddot{\sigma}_x^2 + \ddot{\sigma}_y^2 + (\ddot{\sigma}_z - g)^2}}, \tag{30}$$

from which it can easily be inferred that z is a function of $\boldsymbol{\sigma}$ and derivatives. Finally, use again relations (27) and (28) to obtain

$$\sigma_y - y = \frac{l\ddot{\sigma}_y}{\sqrt{\ddot{\sigma}_x^2 + \ddot{\sigma}_y^2 + (\ddot{\sigma}_z - g)^2}} \tag{31}$$

and

$$\sigma_x - x = \frac{l\ddot{\sigma}_x}{\sqrt{\ddot{\sigma}_x^2 + \ddot{\sigma}_y^2 + (\ddot{\sigma}_z - g)^2}} \tag{32}$$

which completes the proof of our theorem. ■

Note that expressions (27)-(29) are inspired by earlier work on Flatness applied on gantry cranes by Fliess et al. [8], and predating their seminal paper [14]. Remarkably but somewhat unsurprisingly, these dynamic equations do not depend on the mass parameter of the payload, as is known in the related standard dynamic behavior of an undamped pendulum.

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IV. Control Design

The overall structure of the control system, shown in Figure 2, consists of a simple tracking controller, with feedback on the position of the drone based on the well-known ADRC paradigm (see for example [15]). The desired positions of the drone over time are obtained thanks to the developments of the previous section, such that the former are, in turn, functions of the desired trajectories of the payload.

From a control perspective and seen from the drone, the model used is now

$$m\ddot{\mathbf{p}} = \mathbf{F} + mg\mathbf{e}_z + m\mathbf{h}(\mathbf{p}, \dot{\mathbf{p}}, \mathbf{w}, \mathbf{f}_T) \tag{33}$$

where the term $\mathbf{h}(\mathbf{p}, \dot{\mathbf{p}}, \mathbf{w}, \mathbf{f}_T)$ gathers the effect of the unknown tension, unmodelled dynamics and external disturbances. The structure of our controller is expressed as

$$\begin{aligned} \mathbf{F} &= m\ddot{\mathbf{p}}_d - mg\mathbf{e}_z \\ &+ \mathbf{K}_P(\hat{\mathbf{p}} - \mathbf{p}_d) + \mathbf{K}_D(\hat{\dot{\mathbf{p}}} - \dot{\mathbf{p}}_d) - m\hat{\mathbf{h}} \end{aligned} \tag{34}$$

where the two first terms are feedforward terms, and the three next terms come from the linear ADRC controller, with $\hat{\mathbf{p}}, \hat{\dot{\mathbf{p}}}, \hat{\mathbf{h}}$ being estimates provided by an Extended-State Observer (ESO). Time-varying term $\mathbf{p}_d(t)$ is the desired trajectory of the drone. These different elements are detailed below.

A. Extended-State Observer

Using the well-known ESO concept, rewrite the above dynamics (33) as

$$\begin{aligned} \dot{\mathbf{q}}_1 &= \mathbf{q}_2 \\ \dot{\mathbf{q}}_2 &= \frac{1}{m}\mathbf{F} + g\mathbf{e}_z + \mathbf{q}_3 \\ \dot{\mathbf{q}}_3 &= \mathbf{h}(\mathbf{p}, \dot{\mathbf{p}}, \mathbf{w}, \mathbf{f}_T), \end{aligned} \tag{35}$$

with $\mathbf{q}_1 := \mathbf{p}$, $\mathbf{q}_2 := \dot{\mathbf{p}}$ and $\mathbf{q}_3 := \mathbf{h}$ (note that the first time-derivative of $\mathbf{h}$ is assumed to be bounded).
The corresponding ESO reads

$$\begin{aligned} \dot{\hat{\mathbf{q}}}_1 &= \hat{\mathbf{q}}_2 + \mathbf{B}_1(\mathbf{q}_1 - \hat{\mathbf{q}}_1) \\ \dot{\hat{\mathbf{q}}}_2 &= \frac{1}{m}\mathbf{F} + g\mathbf{e}_z + \hat{\mathbf{q}}_3 + \mathbf{B}_2(\mathbf{q}_1 - \hat{\mathbf{q}}_1) \\ \dot{\hat{\mathbf{q}}}_3 &= \mathbf{B}_3(\mathbf{q}_1 - \hat{\mathbf{q}}_1), \end{aligned} \tag{36}$$

where the observer gain matrices $\mathbf{B}_1$, $\mathbf{B}_2$ and $\mathbf{B}_3$ are chosen as diagonal matrices with each term i of the diagonal tuned using the parametrized pole placement technique, ie $b_{1,i} = 3\omega_i$, $b_{2,i} = 3.2\omega_i^2$, $b_{3,i} = \omega_i^3$, and $\omega_i > 0$ is the observer bandwidth.
The related control parameters $\mathbf{K}_P, \mathbf{K}_D$ in (34), again diagonal matrices, are chosen in such a way that $s^2 + \mathbf{K}_D s + \mathbf{K}_P$ is Hurwitz.

B. From drone to payload desired trajectories

As we have seen in the previous section and the differential flatness property of a subset of the overall model, the position of the drone can be obtained by relatively simple functions of the position of the payload and its derivatives. Hence the desired drone position trajectory $\mathbf{p}_d = [x_d, y_d, z_d]^\top$ can also

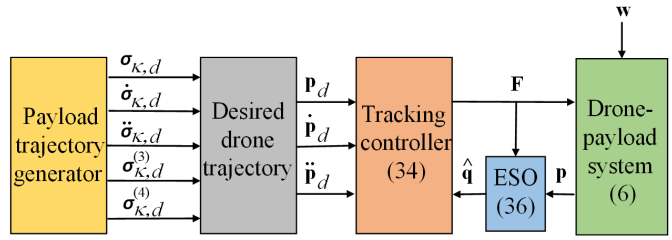

Fig. 2: The structure of the control system

be obtained as functions of desired payload trajectories, ie we have

$$x_d = \sigma_{x,d} - \frac{l\ddot{\sigma}_{x,d}}{\sqrt{\ddot{\sigma}_{x,d}^2 + \ddot{\sigma}_{y,d}^2 + (\ddot{\sigma}_{z,d} - g)^2}} \tag{37}$$

$$y_d = \sigma_{y,d} - \frac{l\ddot{\sigma}_{y,d}}{\sqrt{\ddot{\sigma}_{x,d}^2 + \ddot{\sigma}_{y,d}^2 + (\ddot{\sigma}_{z,d} - g)^2}} \tag{38}$$

$$z_d = \sigma_{z,d} - \frac{l(\ddot{\sigma}_{z,d} - g)}{\sqrt{\ddot{\sigma}_{x,d}^2 + \ddot{\sigma}_{y,d}^2 + (\ddot{\sigma}_{z,d} - g)^2}}, \tag{39}$$

and similarly for the derivatives $\dot{\mathbf{p}}_d$ and $\ddot{\mathbf{p}}_d$ necessary to controller (34).

C. Payload Trajectory Generation

To define the desired payload trajectories, polynomials are used because of their computer efficiency. Furthermore, the desired trajectories should be defined so that the position, velocity, acceleration, and their derivatives up to the fourth order (ie following for example (39), the second-derivative of the drone position depends on the fourth derivative of the payload position) be sufficiently smooth. Therefore, we take ten boundary conditions (ie five initial conditions and five final conditions), meaning a polynomial of order $n = 9$. A polynomial trajectory is expressed as

$$\sigma_{\kappa,d}(t) = a_9 t^9 + a_8 t^8 + a_7 t^7 + \ldots + a_1 t + a_0 \tag{40}$$

where $\kappa = [x, y, z]$. Consider the polynomial coefficients put in vector form

$$\mathbf{a} = [a_9, a_8, \ldots, a_0]^\top \tag{41}$$

and the corresponding time vector

$$\boldsymbol{\lambda}(t) = [t^9, t^8, \ldots, t^2, t, 1], \tag{42}$$

where time variable t is defined on interval $t \in [t_0; t_f]$, $t_0 = 0$ is the starting time of the trajectories and t_f is the time for which the trajectories should reach their final values. The boundary conditions are specified in $\mathbf{c}$ as

$$\begin{aligned} \mathbf{c}(t) = [&\sigma_{\kappa,d}(t_0), \dot{\sigma}_{\kappa,d}(t_0), \ddot{\sigma}_{\kappa,d}(t_0), \sigma_{\kappa,d}(t_0)^{(3)}, \sigma_{\kappa,d}(t_0)^{(4)}, \\ &\sigma_{\kappa,d}(t_f), \dot{\sigma}_{\kappa,d}(t_f), \ddot{\sigma}_{\kappa,d}(t_f), \sigma_{\kappa,d}(t_f)^{(3)}, \sigma_{\kappa,d}(t_f)^{(4)}]^\top \end{aligned} \tag{43}$$

and define $\boldsymbol{\Lambda}$ as a matrix of $\boldsymbol{\lambda}$ and its derivatives evaluated at the initial and final times as

$$\begin{aligned} \boldsymbol{\Lambda}(t) = [&\boldsymbol{\lambda}(t_0), \dot{\boldsymbol{\lambda}}(t_0), \ddot{\boldsymbol{\lambda}}(t_0), \boldsymbol{\lambda}(t_0)^{(3)}, \boldsymbol{\lambda}(t_0)^{(4)}, \\ &\boldsymbol{\lambda}(\mathrm{t}_f), \dot{\boldsymbol{\lambda}}(t_f), \ddot{\boldsymbol{\lambda}}(t_f), \boldsymbol{\lambda}(t_f)^{(3)}, \boldsymbol{\lambda}(t_f)^{(4)}]^\top. \end{aligned} \tag{44}$$

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Then, the coefficients $\mathbf{a}$ can simply be obtained as

$$\mathbf{a} = \mathbf{\Lambda}^{-1}\mathbf{c}, \tag{45}$$

so that we desired trajectories $\sigma_{\kappa,d}(t)$ can be calculated by using (45) in (40).

V. Simulation Study

To illustrate the potential of the proposed control architecture, a simulation in MATLAB/Simulink was implemented for the 3D model of the drone-payload system presented in section II. The parameters considered in the model are assumed to be $m = 70$kg, $m_p = 100$kg, $l = 14.4$m, and $g = 9.81$m/s^2 and inspired by [7]. By using these parameters, we wish to simulate a scenario that is closer to real-world circumstances, enabling us to evaluate the applicability and efficacy of our suggested control structure.

The initial velocity of the drone is considered to be zero. We aim to transfer the payload from a desired initial state to a desired final state in a desired time duration by the attached cable with a length of l from the COG of the drone. Furthermore, to evaluate the robustness of the proposed control, a sudden disturbance at $t = 10s$ is injected into the system. For the simulation test, the disturbance force approximates the effect of the wind with velocity of 10 m/s. For this, we use basic aerodynamic calculations with $\mathbf{w} = \frac{1}{2}c_d\rho\mathbf{A}\cdot\mathbf{v}_r^2$ with reference area $\mathbf{A} = [l_y l_z, l_x l_z, l_x l_y]^\top$ and relative velocity of $\mathbf{v}_r = \dot{\mathbf{p}} - \mathbf{v}_{wind}$. The magnitude of the disturbance force for a wind speed with a magnitude of 10 m/s, drone dimensions $1.6 \times 1.6 \times 0.412$m, $\rho = 1.225$ and $c_d = 0.4$ is shown in Figure 3.

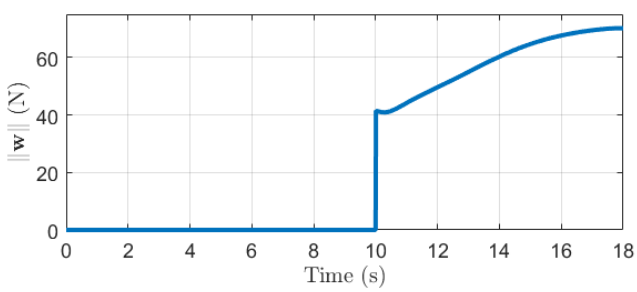

Fig. 3: The magnitude of the injected disturbance force

Figure 4 shows the displacement of the load carried by the drone with the cable in 3D from the initial position of $\mathbf{p}_0^p = [4, 2, -17.4]^\top$ to the desired position of $\mathbf{p}_d^p = [18, 20, -23.4]^\top$ in $t_f - t_0 = 18$s. In Figure 5 the reference trajectories $\sigma_{x,d}, \sigma_{y,d}$ and $\sigma_{z,d}$ start at t_0 from the initial position, and reach the desired position in 18s where the load follows these trajectories. Furthermore, the reference trajectories $\sigma_{x,d}, \sigma_{y,d}$, and $\sigma_{z,d}$ are smooth functions to ensure a minimum number of oscillations for α and β during the transient and after the transfer duration. In particular, the departure and arrival must be sufficiently smooth. It means that $\dot{\sigma}_{\kappa,d} = \ddot{\sigma}_{\kappa,d} = {\sigma_{\kappa,d}}^{(3)} = {\sigma_{\kappa,d}}^{(4)} = 0$ at $t_0 = 0$s and $t_f = 18$s. Furthermore, there is significant resilience against external disturbances since the external disturbances

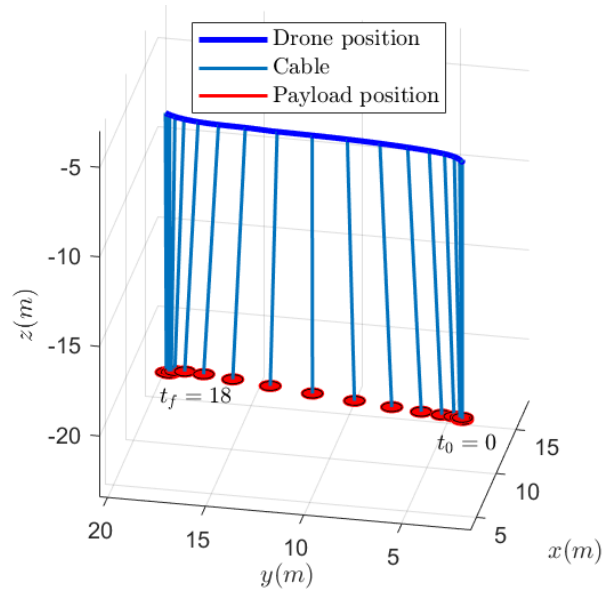

Fig. 4: Payload displacement with the drone with the attached cable

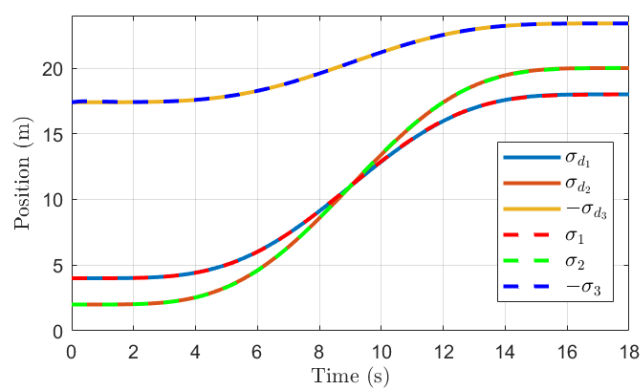

Fig. 5: The position of the payload as the flat output and its reference

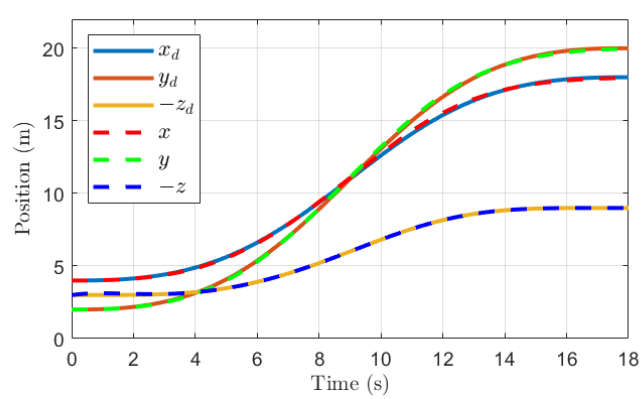

Fig. 6: The position of the drone and its reference

injected into the system at $t = 10s$ are quickly damped down and attenuated. Figure 6 shows the drone's position which

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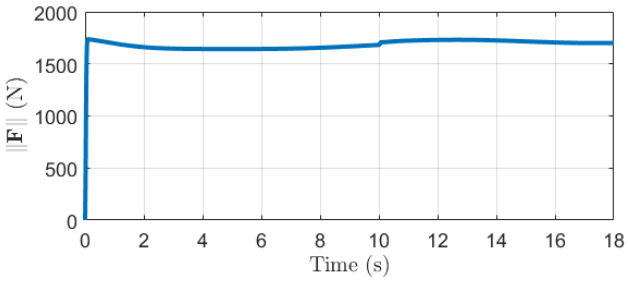

Fig. 7: The magnitude of inertial control force

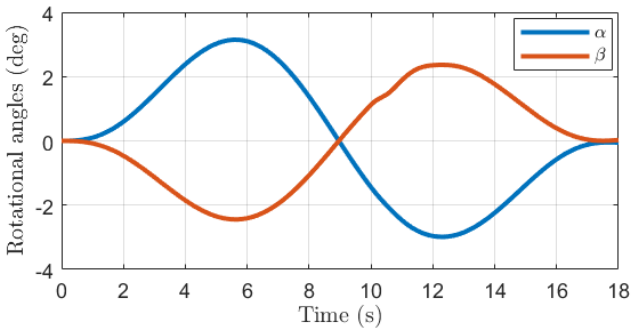

Fig. 8: The rotational angles of the cable

follows its reference.

Figure 7 displays the inertial control force, and the rotational angles of the cable are shown in Figure 8.

It is clear from these simulation results that the slung payload can be smoothly transported from the desired starting position to the desired final position.

VI. Concluding Remarks

In the present work, we looked at drone-with-a-slung-load problem with primary objective the position of the load. With the mass of the load not always known in practice, we proposed a controller architecture whereby the focus was simplicity, so the approach can be used with many different attitude controllers, and did not need the knowledge of the above-mentioned mass nor the deviation angles of the cable. Contrarily to most current work, the core of our approach relies on the use of differential flatness on only a part of the overall drone-payload dynamics. Future work will focus on further robustness characterization and extensions of the proposed approach.

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